Event-Jump State Space Models for Leakage-Safe Financial Disclosure Response Modeling

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Abstract

Financial disclosure modeling is usually treated as feature fusion: a model encodes pre-event market history, appends event text, and predicts post-event returns. This obscures a central scientific question: whether public disclosures act as discontinuous interventions on latent market dynamics. We propose the Event-Jump State Space Model (EJSSM), a neural architecture that separates ordinary no-event evolution from disclosure-induced latent jumps. A recurrent state encoder summarizes pre-event price and volume history, a no-event transition estimates the ordinary continuation, and SEC filing text generates a bounded latent jump before a Gaussian event-response head predicts abnormal return, volatility jump, and volume jump. In the main variant, the jump generator receives disclosure text but not event/control indicator metadata. On a leakage-controlled public-data study with 7,236 SEC 8-K events, 7,236 matched no-event controls, and 2,463 held-out rows from 98 large U.S. firms, a BGE-embedded EJSSM reduces held-out multi-target MSE from 0.0597 for concat fusion to 0.0547. The paired row-bootstrap improvement over concat fusion is 0.0050 with 95% interval [0.0034, 0.0067]. Relative to the deterministic hashing EJSSM, BGE-EJSSM gives a small, non-significant MSE improvement but a significant Gaussian NLL improvement of 0.0709 with 95% interval [0.0357, 0.1153]. Intervention tests show that removing the jump, zeroing disclosure text, or shuffling disclosure text raises event-row MSE by 0.0096, 0.0092, and 0.0128, respectively, with positive paired intervals. The architecture still does not solve abnormal-return ranking; its contribution is event-response state modeling, not trading-signal discovery.

1 Introduction

Disclosure events are not ordinary covariates. A public filing arrives at a known time, changes the information set, and may alter expected return, volatility, liquidity, and factor exposure. Standard multimodal forecasters often mix these effects into a single hidden representation. This can improve prediction, but it gives little structure for distinguishing ordinary market evolution from the discontinuity induced by the event.

This paper studies a stronger architectural hypothesis: disclosure events are sparse jumps in a latent financial state-space model. The hypothesis is not that the model can reliably rank near-term stock returns. Rather, it is that separating ordinary dynamics from event jumps improves leakage-safe modeling of the post-disclosure response distribution and yields a measurable event-specific latent mechanism.

We introduce the Event-Jump State Space Model (EJSSM). Given a pre-event market window X_i, metadata m_i, and disclosure embedding e_i, EJSSM first estimates a no-event latent state $\tilde{s}_i$. It then maps disclosure text to a bounded jump j_i and predicts the event response from $s_i^+ = \mathrm{LN}(\tilde{s}_i + j_i)$. The jump generator excludes event/control indicator metadata; matched no-event controls penalize spurious event deltas on dates where no disclosure is present.

The contributions are:

1. A disclosure-conditioned neural state-space architecture with explicit latent event jumps.
2. A leakage-safe public experiment over SEC 8-K events and matched no-event controls.
3. Evidence that event jumps improve calibrated multi-target event-response regression relative to no-event, text-only, concat-fusion, bottleneck, and operator baselines.

4. Diagnostic evidence that learned jump magnitudes separate real disclosures from matched controls, while abnormal-return rank ordering remains unresolved.

2 Related Work

Classical event studies estimate abnormal returns around information releases [Fama et al., 1969, Brown and Warner, 1985, MacKinlay, 1997]. EJSSM keeps the event-study information boundary but replaces scalar abnormal-return estimation with a neural latent jump model for multiple event-response targets. The architecture is related to state-space and dynamical-system approaches, including Koopman operators [Koopman, 1931], neural differential equations [Chen et al., 2018], structured state-space sequence models [Gu et al., 2022], and selective state-space models [Gu and Dao, 2023]. Marked point-process models study event arrivals and self-excitation [Hawkes, 1971, Mei and Eisner, 2017]; EJSSM instead conditions on observed disclosure timestamps and models the post-event response.

Financial-text research shows that filing language can contain information relevant to prices [Loughran and McDonald, 2011, Cohen et al., 2020, Araci, 2019]. Recent financial forecasting and agent benchmarks study multimodal event forecasting, retrieval, and temporal leakage [Chen et al., 2025, Islam et al., 2023, Jiang et al., 2026, Li et al., 2025]. Causal Transformer and G-Transformer estimate counterfactual outcomes in temporal settings [Melnychuk et al., 2022, Feng et al., 2024]. EJSSM differs by making the event mechanism an explicit bounded jump in latent dynamics rather than a concatenated feature, retrieved context, or unconstrained residual.

3 Method

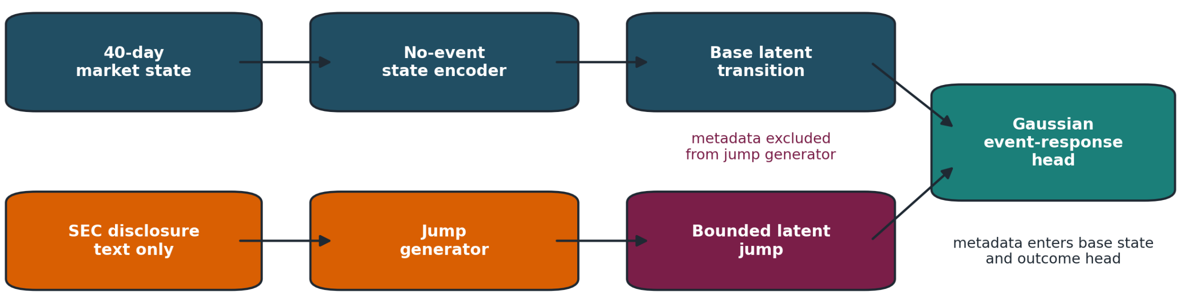

Figure 1: EJSSM separates ordinary market evolution from a disclosure-induced latent jump. Filing text and metadata do not directly overwrite the forecast; they generate a bounded state discontinuity that is added to the no-event latent transition before the Gaussian event-response head.

Inputs and targets. Each sample i has a label start date τ_i, a 40-trading-day pre-event tensor $X_i \in \mathbb{R}^{40\times 8}$, a disclosure embedding e_i, metadata $m_i \in \mathbb{R}^6$, and target vector $y_i \in \mathbb{R}^3$. The targets are five-trading-day SPY-adjusted return, realized-volatility jump, and volume jump. We report both a deterministic hashing encoder and a chunk-pooled BGE-small encoder [Xiao et al., 2023]; the headline BGE variant uses 384-dimensional disclosure embeddings.

No-event state transition. A recurrent encoder maps the pre-event sequence to a latent state:

$$h_i = \mathrm{GRU}_\theta(X_i) + W_m m_i. \tag{1}$$

A bounded no-event gate produces the ordinary continuation:

$$\tilde{s}_i = h_i \odot (1 + 0.1 \tanh(g_\theta(h_i, m_i))). \tag{2}$$

The base head $b_\theta(\tilde{s}_i)$ estimates the response expected without an event jump.

Disclosure jump. The disclosure encoder generates a jump direction and gate from text alone:

$$[u_i, v_i] = r_\theta(e_i), \tag{3}$$

$$j_i = \alpha \tanh(u_i) \odot \sigma(v_i), \tag{4}$$

where $\alpha = 0.20$ in the main experiment. The event state is

$$s_i^+ = \mathrm{LN}(\tilde{s}_i + j_i). \tag{5}$$

The jump is therefore explicit, bounded, and inspectable. For matched no-event controls, the training objective penalizes large event deltas.

Probabilistic event-response head. EJSSM predicts a diagonal Gaussian response distribution:

$$p_\theta(y_i \mid X_i, e_i, m_i) = \mathcal{N}\left(\mu_\theta(s_i^+, m_i), \mathrm{diag}(\exp(\ell_\theta(s_i^+, m_i)))\right). \tag{6}$$

The reported point prediction is μ_θ. The main BGE run combines a target-standardized Huber supervised loss, Gaussian negative log likelihood, control residual suppression, jump sparsity, residual consistency, and a small pairwise ranking regularizer over real disclosure rows:

$$\mathcal{L} = \lambda_y \mathcal{L}_{Huber}(D_y^{-1}(\mu_\theta - y); w_y) + \lambda_n \mathcal{L}_{NLL} + \lambda_c \mathbb{1}_{control} \|\Delta_i\|_2^2 + \lambda_s \|j_i\|_1 + \lambda_v \mathrm{Var}(\Delta_i) + \lambda_r \mathcal{L}_{rank}. \tag{7}$$

where D_y is computed from the training targets and $w_y = (3.0, 1.0, 0.25)$ emphasizes abnormal return relative to volume jumps. The main run uses $\lambda_y = 1.0$, $\lambda_n = 0.05$, $\lambda_c = 0.25$, $\lambda_s = 0.001$, $\lambda_v = 0.05$, and $\lambda_r = 0.15$.

4 Data and Leakage Controls

The experiment uses public SEC EDGAR APIs [U.S. Securities and Exchange Commission, 2026]. The main sample covers 98 usable large U.S. firms, SEC 8-K filings from 2020–2025, and daily public prices. Tickers are mapped to CIKs through the SEC company-tickers file. Filings are read from SEC submissions JSON, including archived submissions, using `acceptanceDateTime` and the primary document URL in EDGAR. The parser stores extracted text, source URL, accession number, accepted timestamp, available timestamp, and text hash.

Daily prices are downloaded from public price endpoints, with SPY used as the market proxy for abnormal returns. The trading calendar is inferred from observed price rows. Event label windows start from the first tradable session after public availability; filings accepted after 16:00 ET or on non-trading days start on the next observed trading day.

Features use only prices strictly before the label start date. The eight numeric features are one-day return, five-day compounded return, 20-day compounded return, 20-day realized volatility, 20-day volume z-score, market return, market-relative return, and close-to-open return. Labels use future returns only as targets. One deterministic same-ticker no-event control is constructed for each filing, excluding dates within a 10-calendar-day blackout window around known events.

Table 1: Main public-data sample.

Artifact	Count
Usable companies	98
SEC 8-K events	7,236
Matched no-event controls	7,236
Feature rows	14,472
Public price rows	159,093
Held-out rows	2,463

5 Experiments

Rows with $\tau_i \leq$ 2023-12-31 are training rows, rows from 2024 are validation rows, and later rows are held out. This yields 9,729 training rows, 2,280 validation rows, and 2,463 held-out rows across 98 held-out tickers and 13 held-out label-start months. Deterministic-hash baselines are trained for 15 epochs; the BGE-EJSSM run is trained for 18 epochs. All runs use AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch size 64, and fixed seed 29.

Baselines include no-event dynamics, text-only MLP, concat fusion, Counterfactual Event Bottleneck Transformer (CEBT), and Disclosure Operator Transformer (DOT). CEBT is an additive stochastic bottleneck baseline. DOT generates a low-rank latent operator from disclosure text. EJSSM differs by modeling the event as a bounded jump in a no-event state-space transition and by predicting a Gaussian response distribution. The BGE-EJSSM variant uses chunk-pooled BGE-small disclosure embeddings and a target-standardized Huber objective with additional abnormal-return weighting; the hashing EJSSM uses the same architecture with deterministic hashed text features.

Confidence intervals use 1,000 percentile bootstrap resamples for standalone metrics and 5,000 row-paired bootstrap resamples for paired MSE comparisons. Ticker-clustered intervals resample tickers with replacement. Rank IC is Spearman correlation between predicted and realized abnormal return over held-out rows. We also evaluate three EJSSM interventions on the same checkpoint: setting $j_i = 0$, replacing disclosure embeddings with zero vectors, and shuffling disclosure embeddings across held-out rows.

Table 2: Held-out results. MSE is over the three event-response targets. Intervals are 95% ticker-clustered bootstrap intervals.

Model	MSE	MSE CI	Rank IC	Rank IC CI
No-event	0.0696	[0.0519, 0.0936]	-0.0411	[-0.0840, 0.0016]
Text-only	0.0643	[0.0495, 0.0844]	-0.0009	[-0.0456, 0.0434]
Concat fusion	0.0597	[0.0442, 0.0817]	-0.0182	[-0.0597, 0.0267]
CEBT	0.0660	[0.0488, 0.0898]	**0.0463**	[0.0052, 0.0893]
DOT	0.0596	[0.0449, 0.0804]	-0.0131	[-0.0510, 0.0291]
EJSSM-hash	0.0551	[0.0412, 0.0748]	0.0027	[-0.0410, 0.0431]
EJSSM-BGE	**0.0547**	[0.0395, 0.0764]	0.0099	[-0.0329, 0.0556]

Table 3: Paired MSE comparisons against EJSSM. Positive values mean EJSSM has lower MSE.

Baseline	Baseline–EJSSM MSE	95% paired CI
No-event	0.01493	[0.01218, 0.01814]
Text-only	0.00963	[0.00746, 0.01187]
Concat fusion	0.00504	[0.00339, 0.00672]
CEBT	0.01134	[0.00899, 0.01395]
DOT	0.00494	[0.00296, 0.00695]
EJSSM-hash	0.00044	[-0.00248, 0.00304]

Table 4: EJSSM diagnostic metrics on the held-out split.

Diagnostic	Value
Gaussian NLL	-2.2839
Observed 80% interval coverage	0.8921
Observed 95% interval coverage	0.9594
Mean predictive standard deviation	0.1362
Mean latent jump, real 8-Ks	0.0485
Mean latent jump, controls	0.0426
Mean event delta, real 8-Ks	0.0403
Mean event delta, controls	0.0125

Table 5: EJSSM intervention tests on held-out rows. Lower MSE and NLL are better. Jump AUC tests whether latent jump magnitude separates real 8-K rows from matched no-event controls.

Variant	All MSE	Event MSE	Event NLL	Rank IC	Jump AUC
Full EJSSM-BGE	**0.0547**	**0.0831**	**-2.2215**	0.0099	**0.7236**
No jump	0.0600	0.0926	-2.1829	-0.0028	0.5000
Zero disclosure	0.0591	0.0922	-2.1795	-0.0031	0.5000
Shuffled disclosure	0.0621	0.0959	-2.1602	-0.0209	0.5254

Table 6: Paired event-row MSE stress tests. Positive values mean full EJSSM has lower MSE than the intervention.

Intervention	MSE increase	95% paired CI	95% ticker-clustered CI
No jump	0.0096	[0.0064, 0.0131]	[0.0057, 0.0141]
Zero disclosure	0.0092	[0.0062, 0.0124]	[0.0056, 0.0133]
Shuffled disclosure	0.0128	[0.0088, 0.0170]	[0.0084, 0.0178]

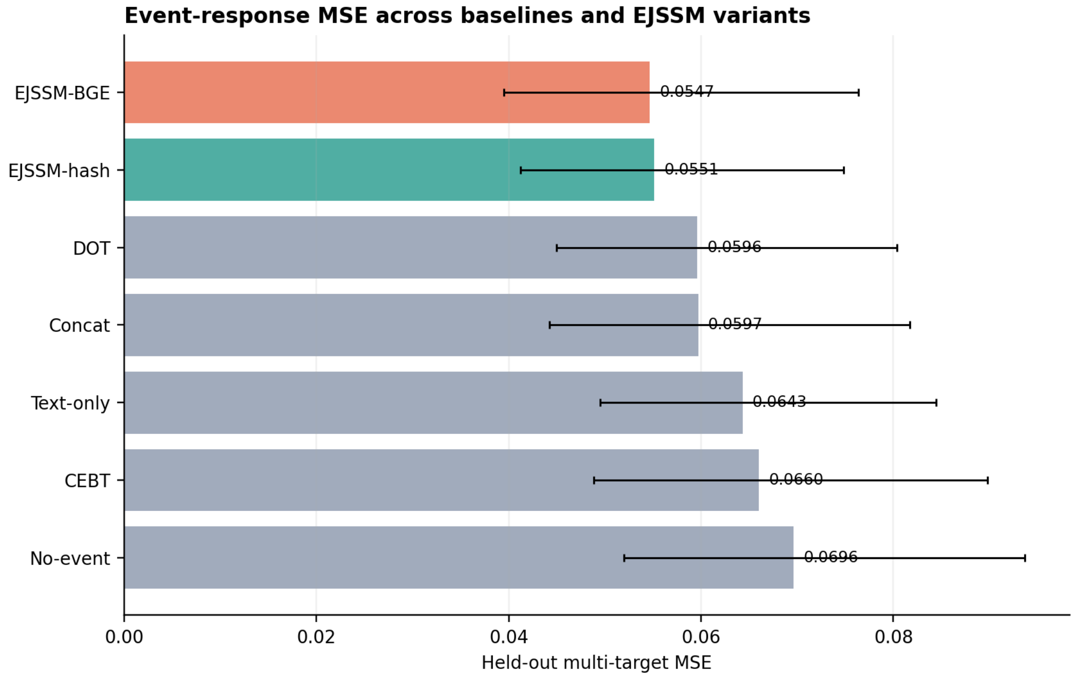

Figure 2: Held-out MSE with ticker-clustered bootstrap intervals. EJSSM has the best point estimate and improves significantly over the strongest fusion baselines under paired row resampling.

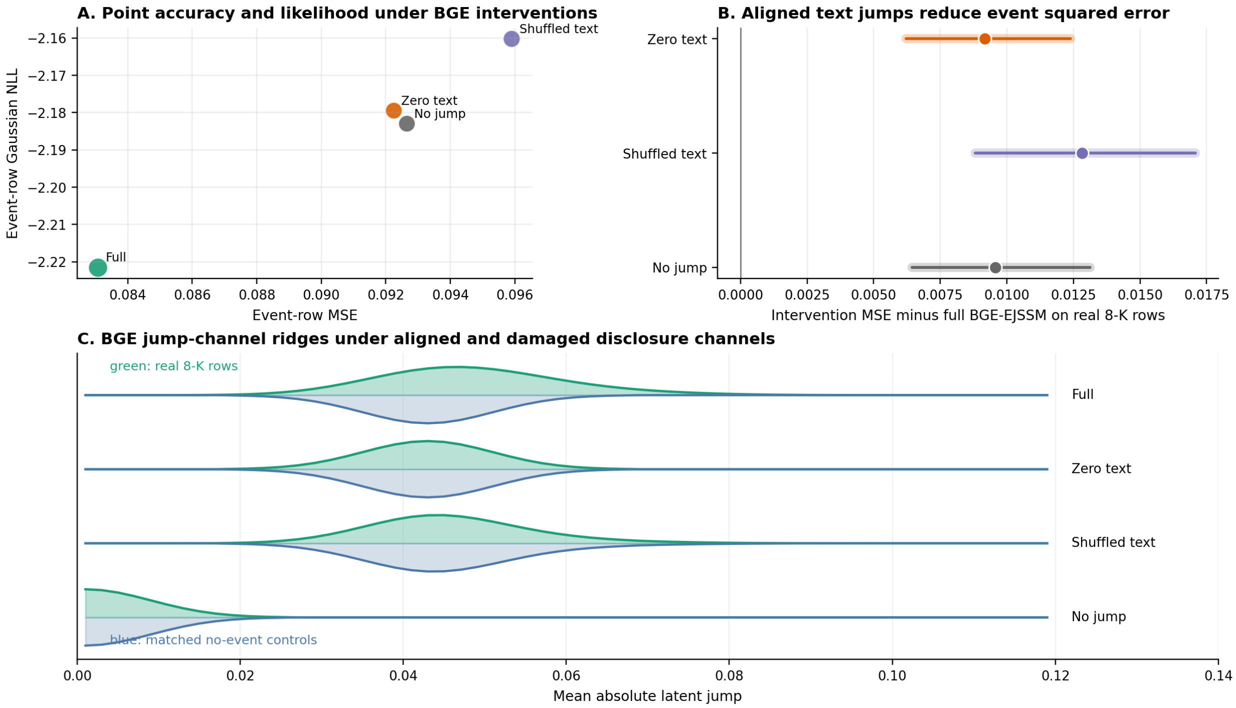

Figure 3: Intervention stress test. Full EJSSM has the lowest event-row MSE and Gaussian NLL among the tested BGE interventions. The ridge panel shows that aligned disclosure text creates an event/control jump separation, whereas zeroing the disclosure channel or removing the jump collapses this separation and shuffling text weakens it.

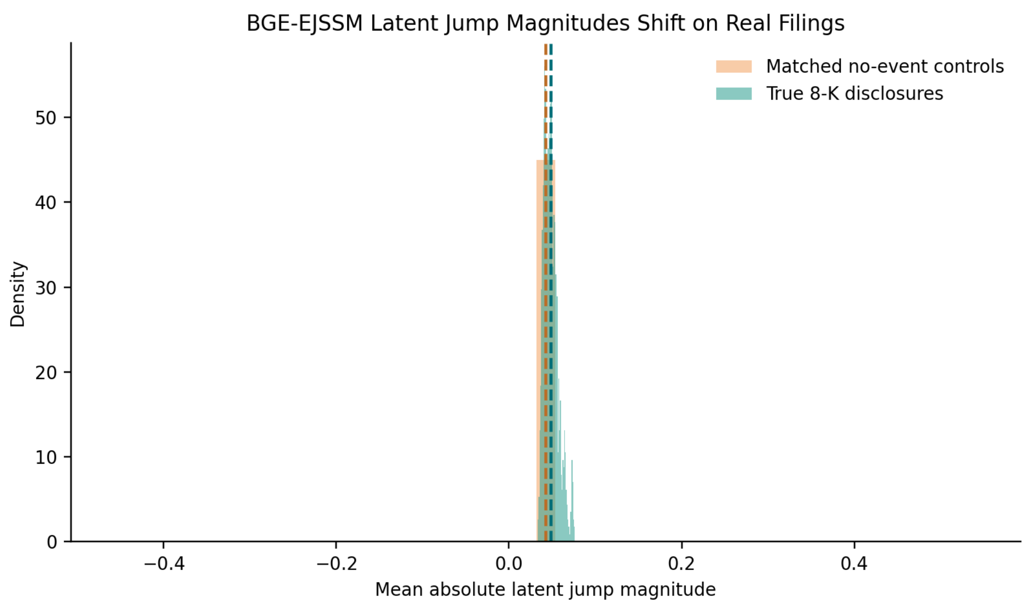

Figure 4: Latent jump magnitudes for real disclosures and matched controls. Real 8-K events shift the learned jump distribution away from the low-magnitude control regime.

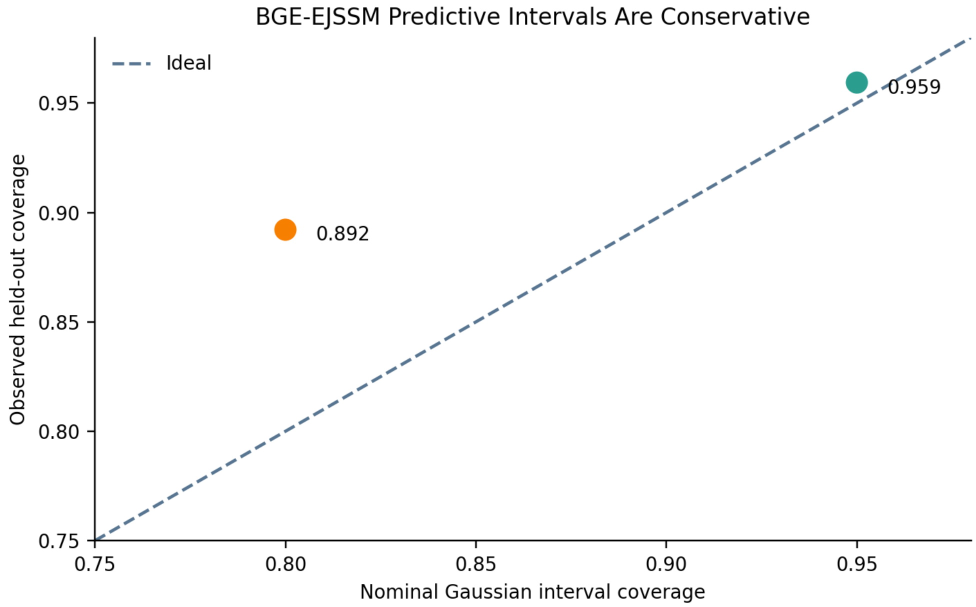

Figure 5: Held-out Gaussian interval calibration. EJSSM is conservative at 80% coverage and close to nominal at 95% coverage.

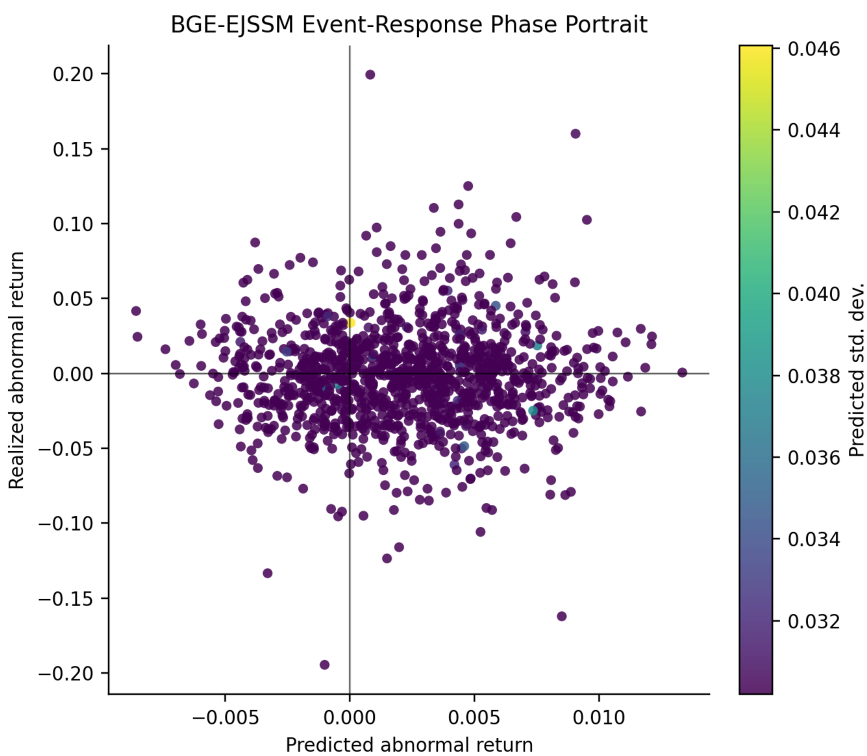

Figure 6: Event-response phase portrait for held-out real disclosures, colored by predicted abnormal-return uncertainty. The model improves response regression while still leaving a wide realized-return dispersion, which explains the weak rank-IC result.

6 Findings

Latent jumps improve event-response regression. BGE-EJSSM has held-out MSE 0.0547, compared with 0.0597 for concat fusion and 0.0596 for DOT. The paired MSE improvement over concat fusion is 0.0050 with interval [0.0034, 0.0067], and the improvement over DOT is 0.0049 with interval [0.0030, 0.0070]. Relative to the hashing EJSSM, the MSE improvement is small and not significant: 0.0004 with interval [-0.0025, 0.0030].

The learned jump channel is active on real disclosures. Mean absolute latent jump magnitude is 0.0485 for real 8-K rows and 0.0426 for matched no-event controls. Mean absolute event delta is 0.0403 for real 8-K rows and 0.0125 for controls. These gaps support the interpretation that the model uses its jump channel differently when a public filing is present.

Interventions support a narrow mechanism claim. Setting the latent jump to zero raises event-row MSE by 0.0096 with interval [0.0064, 0.0131]. Zeroing disclosure text raises event-row MSE by 0.0092 with interval [0.0062, 0.0124], and shuffling disclosure text raises it by 0.0128 with interval [0.0088, 0.0170]. These results show that aligned disclosure-conditioned jumps help squared-error response prediction. They do not show that the jump norm estimates realized event magnitude: in held-out real 8-K rows, jump magnitude is not positively rank-correlated with absolute abnormal return or volatility jump.

The probabilistic head is useful but imperfect. The 95% Gaussian interval has observed coverage 0.9594, slightly above nominal. The 80% interval has coverage 0.8921, indicating conservative uncertainty at narrower intervals. BGE-EJSSM substantially improves Gaussian NLL over the hashing EJSSM: the paired NLL improvement is 0.0709 with interval [0.0357, 0.1153]. This is the clearest gain from replacing hashing with an open pretrained disclosure encoder.

EJSSM does not solve event ranking. The abnormal-return rank IC for BGE-EJSSM is 0.0099 with ticker-clustered interval [-0.0329, 0.0556]. CEBT remains the only tested model with positive clustered rank-IC interval. Thus, the main EJSSM result is response regression, distributional calibration, and intervention sensitivity, not cross-sectional return ranking.

7 Reproducibility

The accompanying artifact records the exact configuration, random seed, processed event metadata, text hashes, price rows, feature metadata, model checkpoints, predictions, table CSVs, and figure outputs used for the reported results. The implementation includes tests for timestamp gating, after-hours label starts, feature leakage rejection, control construction, deterministic cached tensors, finite forward losses, probabilistic loss integration, and evaluation leakage guards. Labels are computed only from observed future price windows and are never included in model inputs.

8 Limitations

The study has several limitations. BGE-small is an open general-purpose encoder rather than a financial-domain encoder, and SEC filings are represented by chunk mean pooling rather than a trained long-document model. The jump generator excludes event/control metadata, but controls still use zero event embeddings, so stronger placebo-text controls and within-time shuffles remain necessary. The intervention results show that point MSE, likelihood, and ranking can move in different directions; EJSSM should therefore be interpreted as a state-structured response regressor rather than a complete probabilistic or alpha-ranking model. The price source is public daily data rather than CRSP or intraday trades and quotes. The universe is large-cap biased and may contain survivorship effects, ticker-history issues, and 8-K item-mix sensitivity. Abnormal returns are SPY-adjusted rather than factor-model residuals. EJSSM improves event-response MSE but does not produce a statistically reliable abnormal-return ranking signal. The jump interpretation is architectural and predictive rather than structural causal identification.

9 Ethics and Use

This paper studies event representations using public filings and public daily prices. The results do not constitute trading recommendations. The artifact is intended to support leakage auditing, independent reproduction, and falsification of the empirical claims.

10 Conclusion

EJSSM models disclosures as bounded jumps in latent market dynamics. On a real SEC 8-K experiment, this state-space formulation improves held-out multi-target event-response MSE over concat fusion, DOT, CEBT, text-only, and no-event baselines. Replacing deterministic hashed text features with chunk-pooled BGE embeddings yields the strongest overall MSE and a significant Gaussian NLL improvement over the hashing EJSSM. No-jump, zero-disclosure, and shuffled-disclosure interventions all worsen event-row MSE, which supports the architectural value of aligned disclosure-conditioned jumps. The architecture does not produce a statistically reliable abnormal-return rank signal, so the result should be read as a contribution to event-response state modeling rather than trading-signal discovery.

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